

PROJECT PROPOSAL

RESTAURANT MANAGEMENT SYSTEM

A Web-Based Application for Restaurant Operations and Billing

PROJECT INFORMATION

Submitted By	Rifat Islam
Trainee ID	1297436
Batch	PWAD/USSL-M/73/01
Institute	IsDB-BISEW
Class Instructor	Syed Ziaul Habib
Consultation / Supervisor	Md. Moshaidul Islam
Submission Date	03 November 2026

Submitted to IsDB-BISEW

1. Project Overview

The Restaurant Management System is a simple web-based application designed to help a single restaurant manage its food menu, categories, customer information, orders, billing and basic sales records from one place. The system replaces several manual tasks with a centralized application that can be used by an administrator and cashier.

The project will be developed using Core PHP and MySQL for the server-side operations, with HTML, CSS, Bootstrap 5 and basic JavaScript for the user interface. The design and feature set are intentionally limited to keep the project achievable for a beginner while still demonstrating the main functions of a restaurant point-of-sale workflow.

2. Project Objectives

- To maintain food categories and menu items in a MySQL database.
- To store basic customer information when required.
- To create restaurant orders by selecting food items and quantities.
- To calculate order totals automatically and record the payment method.
- To generate a printable invoice for each completed order.
- To view simple daily and monthly sales summaries.
- To provide separate access permissions for the administrator and cashier.

3. Project Scope

3.1 Included in the Final Version

- Admin and cashier login with session-based access control.
- Dashboard showing basic totals such as today's sales, order count and food item count.
- Category and food management (add, view, update and delete).
- Customer records with name, phone and optional email.
- Order creation with multiple food items and quantities.
- Automatic bill calculation, payment method recording and invoice printing.
- Order history and basic date-wise sales reports.
- Restaurant name, phone and address stored in a configuration file and shown on printed invoices.

3.2 Not Included in This Version

The following features are deliberately excluded to keep the final project simple and deliverable: kitchen display, table reservation, ingredient-level inventory, supplier management, employee payroll, online customer ordering, delivery tracking, online payment gateway integration, multi-branch support, mobile applications and advanced chart-based analytics.

> **Scope decision**
> This is a single-restaurant system. It is not a multi-tenant platform; one installation is configured for one restaurant. The excluded features can be considered as future enhancements after the basic version is completed and tested.

4. User Roles and Permissions

The system has two user roles. The role is assigned by the administrator and checked after login.

User Role	Main Responsibilities
Administrator	Manage user accounts, food categories, food items, customers, orders, sales reports.
Cashier	Access the dashboard, create customer records, create orders, record payment methods, view order history and print invoices.

5. Main System Modules

5.1 Login and User Management

Secure login and logout. The administrator can create or deactivate cashier accounts. Passwords are stored as hashes, not plain text.

5.2 Dashboard

Displays a small set of useful figures, such as today's sales, today's completed orders, total food items and total customers. Information is shown using summary cards and simple tables.

5.3 Category Management

Allows the administrator to add, edit, view and delete food categories such as Burger, Pizza, Rice and Drinks. A category cannot be deleted while food items are linked to it unless those items are reassigned or removed first.

5.4 Food Management

Allows the administrator to maintain the food name, category, price, description, optional image and availability status. Unavailable food items cannot be added to a new order.

5.5 Customer Management

Stores customer name, phone number and optional email address. A customer record is optional for a walk-in order, so the cashier can create an order without registering every customer.

5.6 Order and Billing Management

Allows the cashier to create an order, select multiple food items, enter quantities, review the calculated total and save the order. The system creates an order number and stores each ordered item separately.

5.7 Invoice and Payment Recording

Records the payment method selected at checkout, such as Cash, Card, bKash or Nagad. The system prints a simple invoice containing restaurant information, order number, date, items, quantities, prices and total amount. Payment methods are recorded manually; no payment gateway is connected.

5.8 Sales Report

Provides a basic order list and sales summary for a selected date or date range. Reports can show order number, date, cashier, payment method and total amount. No chart library is required.

6. System Workflow

The following workflow describes how a normal restaurant sale is completed in the system:

1. The administrator or cashier opens the application and logs in with a valid account.
2. The dashboard opens according to the user role.
3. The administrator maintains food categories and food items before sales begin.
4. The cashier optionally selects an existing customer or enters a new customer record.
5. The cashier starts a new order and selects one or more available food items.
6. The cashier enters the quantity for each item. The system calculates each subtotal and the order total.
7. The cashier confirms the order and selects the payment method.
8. The system saves the order and its order items, then displays the invoice.
9. The cashier prints the invoice and the completed order becomes available in the order history and sales report.

6.1 Basic Billing Example

Food Item	Quantity	Unit Price	Subtotal
Burger	2	BDT 250.00	BDT 500.00
Pizza	1	BDT 450.00	BDT 450.00
Soft Drink	2	BDT 50.00	BDT 100.00
TOTAL			BDT 1,050.00

Calculation rule: Subtotal = Quantity × Unit Price. Order Total = Sum of all item subtotals. The system uses the food price saved in the database at the time the order is created. That price is copied into the order item so that an old invoice remains correct even if the menu price changes later.

6.2 Important Business Rules

- Only logged-in users can access protected pages.
- Only available food items can be added to a new order.
- Quantity must be a positive whole number.
- The server recalculates prices and totals; it does not trust totals sent by the browser.
- A completed order must have at least one order item.
- A walk-in order may have no customer record.
- Cancelled orders must not be counted as completed sales.

7. Technology Stack and Requirements

Technology	Purpose
HTML5	Page structure and forms.
CSS3	Basic visual styling.
Bootstrap 5	Responsive layout, navigation, tables, forms and buttons.
JavaScript	Simple browser-side interactions and form usability.
Core PHP	Login, session handling, CRUD operations, order processing and reports.
MySQL	Stores user, category, food, customer and order data.
Apache / XAMPP	Local development server and PHP execution environment.
phpMyAdmin	Database creation and administration during development.

7.1 Development Environment

- A computer with Windows, Linux or macOS.
- XAMPP or another PHP and MySQL local development environment.
- A modern web browser such as Chrome, Edge or Firefox.
- A code editor such as Visual Studio Code.

7.2 Proposed Project Folder Structure

```
restaurant_management/
|-- assets/
|   |-- css/style.css
|   |-- js/script.js
|   |-- images/
|-- config/
|   |-- db.php
|   |-- restaurant.php
|-- includes/
|   |-- auth.php
|   |-- header.php
|   |-- sidebar.php
|   |-- footer.php
|-- admin/
|   |-- dashboard.php
|   |-- users.php
|   |-- categories.php
|   |-- foods.php
|   |-- customers.php
|   |-- orders.php
|   |-- reports.php
|-- invoices/
|   |-- print.php
|-- index.php
|-- login.php
|-- logout.php
```

Folder names may be adjusted during coding, but keeping configuration, shared layout files, modules and static assets separate will make the project easier to understand and maintain.

8. Database Design

Database name: restaurant_db. The database contains six tables. Primary keys uniquely identify records. Foreign keys connect related records and help maintain data consistency.

No.	Table Name	Purpose
1	users	Stores administrator and cashier login accounts.
2	categories	Stores food categories.
3	foods	Stores menu items and their category, price and availability.
4	customers	Stores optional customer contact information.
5	orders	Stores each sale, cashier, optional customer, total and payment method.
6	order_items	Stores the individual food items and quantities belonging to an order.

8.1 Entity Relationship Diagram (ERD)

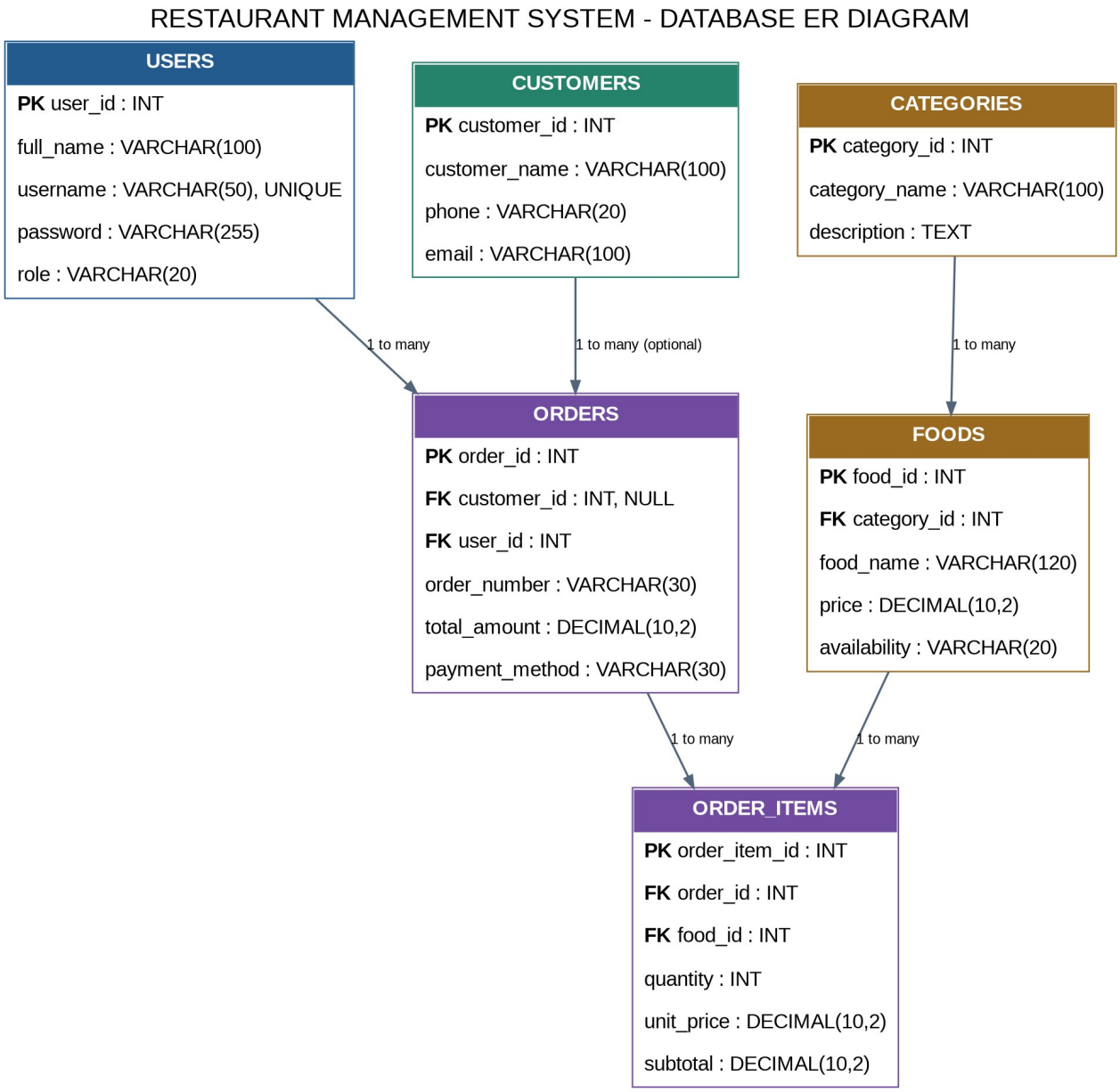


Figure 1. Database entities and their one-to-many relationships.

Relationship summary

One category can contain many foods. One customer can have many orders, while customer_id may be empty for a walk-in sale. One user can process many orders. One order has many order items. One food can appear in many order items. Restaurant details are stored in a simple PHP configuration file and are not a database table.

9. Database Table Structure

The following data dictionary lists the recommended fields for the final beginner-level version. “PK” means Primary Key and “FK” means Foreign Key.

9.1 users

Field	Data Type	Key / Rule	Description
user_id	INT	PK, AUTO_INCREMENT	Unique user ID.
full_name	VARCHAR(100)	NOT NULL	User full name.
username	VARCHAR(50)	UNIQUE, NOT NULL	Login username.
password	VARCHAR(255)	NOT NULL	Password hash generated by PHP.
role	VARCHAR(20)	NOT NULL	admin or cashier.
status	TINYINT	DEFAULT 1	1 = active, 0 = inactive.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation date.

9.2 categories

Field	Data Type	Key / Rule	Description
category_id	INT	PK, AUTO_INCREMENT	Unique category ID.
category_name	VARCHAR(100)	UNIQUE, NOT NULL	Name of the food category.
description	TEXT	NULL allowed	Optional category description.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation date.

9.3 foods

Field	Data Type	Key / Rule	Description
food_id	INT	PK, AUTO_INCREMENT	Unique food ID.
category_id	INT	FK, NOT NULL	References categories.category_id.
food_name	VARCHAR(120)	NOT NULL	Food name.
price	DECIMAL(10,2)	NOT NULL	Current selling price.
description	TEXT	NULL allowed	Optional food description.
image	VARCHAR(255)	NULL allowed	Optional image file path.
availability	VARCHAR(20)	DEFAULT Available	Available or Unavailable.

9. Database Table Structure (Continued)

9.4 customers

Field	Data Type	Key / Rule	Description
customer_id	INT	PK, AUTO_INCREMENT	Unique customer ID.
customer_name	VARCHAR(100)	NOT NULL	Customer name.
phone	VARCHAR(20)	NULL allowed	Contact number.
email	VARCHAR(100)	NULL allowed	Optional email address.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation date.

9.5 orders

Field	Data Type	Key / Rule	Description
order_id	INT	PK, AUTO_INCREMENT	Unique internal order ID.
order_number	VARCHAR(30)	UNIQUE, NOT NULL	Human-readable invoice/order number.
customer_id	INT	FK, NULL allowed	References customers.customer_id; empty for walk-in.
user_id	INT	FK, NOT NULL	References users.user_id (cashier).
total_amount	DECIMAL(10,2)	NOT NULL	Total of all order item subtotals.
payment_method	VARCHAR(30)	NOT NULL	Cash, Card, bKash or Nagad.
order_status	VARCHAR(20)	DEFAULT Completed	Completed or Cancelled.
order_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date and time of the order.

9.6 order_items

Field	Data Type	Key / Rule	Description
order_item_id	INT	PK, AUTO_INCREMENT	Unique order line ID.
order_id	INT	FK, NOT NULL	References orders.order_id.
food_id	INT	FK, NOT NULL	References foods.food_id.
quantity	INT	NOT NULL	Number of units ordered; must be greater than zero.
unit_price	DECIMAL(10,2)	NOT NULL	Food price copied at the time of sale.
subtotal	DECIMAL(10,2)	NOT NULL	Quantity multiplied by unit_price.

10. Main Application Pages

Page	Main Purpose
Login	Checks username and password and starts a user session.
Dashboard	Shows key daily totals and navigation links.
User Management	Allows the administrator to add or deactivate cashier accounts.
Category Management	Lists categories and provides add, edit and delete actions.
Food Management	Lists menu items and provides add, edit, delete and availability actions.
Customer Management	Lists customer records and provides add, edit and delete actions.
Create Order	Selects food items and quantities, calculates the bill and records payment method.
Order History	Shows previous orders and allows invoice viewing or printing.
Sales Report	Shows order totals for a selected date or date range.
Print Invoice	Displays a printer-friendly receipt for a selected order.

11. Basic Reports

- Today's sales total.
- Number of completed orders for a selected date.
- Sales summary by date range or month.
- Order-wise details including order number, order date, cashier, payment method and total amount.

Reporting limitation

The final version uses simple tables and totals. Interactive charts, forecasting and advanced analytics are outside the scope of this beginner project.

12. Functional Requirements

- The system shall allow an active administrator or cashier to log in and log out.
- The system shall restrict pages and actions according to the user role.
- The administrator shall be able to manage categories and food items.
- The system shall allow customer records to be created, updated and viewed.
- The cashier shall be able to create an order containing multiple food items.
- The system shall calculate item subtotals and the order total automatically.
- The system shall save order and order item records together.
- The system shall record a payment method and generate a printable invoice.
- The system shall display order history and date-based sales totals.

13. Security and Data Validation

- Store passwords using PHP password_hash() and verify them using password_verify(). Never store plain-text passwords.
- Use prepared statements (PDO or MySQLi) for all database queries that include user input.
- Validate all form data on the server, even when browser-side validation is also used.
- Use session checks on protected pages and verify the user role before sensitive actions.
- Read food prices from the database during order processing; do not trust prices or totals submitted by the browser.
- Use a database transaction when saving an order and its order items so partial orders are not left behind if saving fails.
- Validate uploaded food images by file type and size, and generate safe file names.
- Back up the MySQL database regularly during testing and before deployment.

14. Testing Plan

Test Area	Test Example	Expected Result
Login	Enter correct and incorrect credentials.	Valid user can log in; invalid user is rejected.
Role Access	Open an admin-only page as cashier.	Access is denied or the user is redirected.
Category CRUD	Add, edit and delete a category.	Changes are saved and displayed correctly.
Food CRUD	Add food with price and category.	Food is stored and linked to the correct category.
Order Calculation	Add multiple foods and quantities.	Each subtotal and total are calculated correctly.
Order Saving	Save an order with several items.	Order and all order items are saved together.
Invoice	Open and print a completed order.	Invoice details and totals match the saved order.
Sales Report	Select a date range.	Only matching non-cancelled orders are included.

15. Development Plan

Phase	Development Task
1	Create project folders and the MySQL database.
2	Build database connection, login/logout and the Bootstrap 5 dashboard.
3	Develop category, food and customer CRUD modules.
4	Develop order creation, order items and bill calculation.
5	Develop invoice printing, order history and basic sales reports.
6	Test all modules, fix errors and prepare the final demonstration.

16. Project Deliverables

- A working PHP and MySQL web application for one restaurant.
- A MySQL database named restaurant_db with six related tables, plus a configuration file for restaurant details.
- Admin and cashier login with role-based page access.
- Food category, food item and customer management pages.
- Order creation, automatic bill calculation, payment method recording and invoice printing.
- Order history and basic sales reports.
- A final project report and a demonstration of the completed application.

17. Future Enhancements (Not Part of the Final Scope)

After the basic system is completed and stable, the following modules may be added as a separate upgrade:

- Ingredient-level inventory and low-stock notifications.
- Kitchen order display and preparation status.
- Table management and table reservations.
- Supplier and purchase management.
- Employee attendance and payroll.
- Online ordering and delivery management.
- Payment gateway integration and digital receipts.
- Multi-branch and multi-restaurant support.

18. Conclusion

The Restaurant Management System provides a practical and manageable solution for the basic daily operations of a single restaurant. It focuses on the essential workflow: maintaining the menu, recording customers when needed, creating orders, calculating bills, recording payment methods, printing invoices and reviewing sales. The limited scope makes the project suitable for beginner-level development and academic submission while leaving a clear path for future improvements.

Final scope confirmation

The final submission version contains the core modules documented in this report. Kitchen management, reservations, inventory, payroll, delivery, online payment gateways, multi-branch support and advanced analytics are excluded from this version.

Suggested Project Demonstration

During the final presentation, demonstrate the application in this order: log in as administrator, add a category, add a food item, log in as cashier, create a customer, place an order with two or more food items, show the automatic total, select a payment method, save the order, print the invoice and open the sales report.